

A15536W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics, and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMIC ANALYSIS

OUTLINE ANSWERS

TRINITY TERM 2020

Thursday 04 June, 09:30–12:30

Please start the answer to each question on a separate page.

There are 6 questions in this paper.

*Answer **four** questions.*

For multipart questions, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

1. (a) (i)

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

$\det A = -1$, so yes, A is invertible.

Characteristic polynomial: $-(1 - \lambda)\lambda - 1 = \lambda^2 - \lambda - 1 = 0$. Thus eigenvalues are

$$\lambda_{1,2} = \frac{1 \pm \sqrt{5}}{2}$$

Then, eigenvectors are:

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1 + \sqrt{5}}{2} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \frac{1 + \sqrt{5}}{2} \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1 - \sqrt{5}}{2} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \frac{1 - \sqrt{5}}{2} \\ 1 \end{pmatrix}$$

(ii) Now,

$$\begin{pmatrix} F_{2020} \\ F_{2019} \end{pmatrix} = A^{2018} \cdot \begin{pmatrix} F_2 \\ F_1 \end{pmatrix}$$

Note that $A^{2018} = SD^{2018}S^{-1}$, where D is a diagonal matrix of eigenvalues and S is a matrix of eigenvectors. Thus,

$$\begin{aligned} A^{2018} &= \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1^{2018} & 0 \\ 0 & \lambda_2^{2018} \end{pmatrix} \begin{pmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{pmatrix} \\ &= \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} \lambda_1^{2019} - \lambda_2^{2019} & -\lambda_2 \lambda_1^{2019} + \lambda_1 \lambda_2^{2019} \\ \lambda_1^{2018} - \lambda_2^{2018} & -\lambda_2 \lambda_1^{2018} + \lambda_1 \lambda_2^{2018} \end{pmatrix} \end{aligned}$$

Then using $\lambda_1 = 1 - \lambda_2$ we get

$$A^{2018} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{\lambda_1^{2020} - \lambda_2^{2020}}{\lambda_1 - \lambda_2} \\ \frac{\lambda_1^{2019} - \lambda_2^{2019}}{\lambda_1 - \lambda_2} \end{pmatrix}$$

So

$$F_{2020} = \frac{\left(\frac{1 + \sqrt{5}}{2}\right)^{2020} - \left(\frac{1 - \sqrt{5}}{2}\right)^{2020}}{\sqrt{5}}$$

(b) (i) Compute Jacobian:

$$J_{xy} = \begin{pmatrix} 5x^4 + 2t & 2sy \\ 2y^2 + s^2 & 4xy \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix}$$

Jacobian is not invertible, so IFT does not apply.

(ii) In this case Jacobian is

$$J_{xy} = \begin{pmatrix} 5x^4 + 2t & 2sy \\ 2y^2 + s^2 & 4xy \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 3 & 0 \end{pmatrix}$$

which is invertible with

$$J_{xy}^{-1} = \begin{pmatrix} 2 & -2 \\ 3 & 0 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 0 & 2 \\ -3 & 2 \end{pmatrix}$$

Then

$$J_{st} = \begin{pmatrix} y^2 + 2s & 2x \\ 2sx & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

Thus,

$$\begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{pmatrix} = -J_{xy}^{-1} \cdot J_{st} = -\frac{1}{6} \begin{pmatrix} 0 & 2 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1/3 \\ -1/2 & -1/3 \end{pmatrix}$$

2. (a) (i) f is concave: $f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$ for all $\lambda \in (0, 1)$
(ii) Write $m(x) = \min\{f(x), g(x)\}$. Then $f(x) \geq m(x)$ and $g(x) \geq m(x)$, so:

$$\begin{aligned} f(\lambda a + (1 - \lambda)b) &\geq \lambda f(a) + (1 - \lambda)f(b) \\ &\geq \lambda m(a) + (1 - \lambda)m(b) \\ \text{and similarly } g(\lambda a + (1 - \lambda)b) &\geq \lambda g(a) + (1 - \lambda)g(b) \\ &\geq \lambda m(a) + (1 - \lambda)m(b) \\ \text{therefore } m(\lambda a + (1 - \lambda)b) &\geq \lambda m(a) + (1 - \lambda)m(b) \end{aligned}$$

(iii) Hessians:

$$H_f = \begin{pmatrix} -1/x^2 & 0 \\ 0 & -3/y^2 \end{pmatrix}$$

$\det > 0$, $\text{tr} < 0$, so H_f is negative definite and f is concave.

(If the answer uses “sum of concave functions is concave”, then it should note that $\ln(x)$ is concave as a function of x and y .)

$$H_g = \begin{pmatrix} 0 & 3y^2 \\ 3y^2 & 6xy \end{pmatrix}$$

$\text{tr} > 0$ is possible, so H_g is not negative-semidefinite and g is not concave.

(note that g is an increasing transform of f)

$$H_h = \begin{pmatrix} -2 & 5 \\ 5 & -6 \end{pmatrix}$$

$\text{tr} = -8 < 0$ but $\det = 12 - 25 < 0$, so H_h is not negative semi-definite and h is not concave. [35%]

- (b) (i) Marginal utility tends to infinity as x or y tend to 0.
(ii) Lagrangian $\mathcal{L} = \ln x + 3 \ln y + \lambda(20 - x - 4y) + \mu(20 - 4x - y)$
K-T FOCs:

$$x^{-1} = \lambda + 4\mu \quad (1)$$

$$3y^{-1} = 4\lambda + \mu \quad (2)$$

$$\lambda \geq 0, \quad x + 4y \leq 20, \quad \lambda(20 - x - 4y) = 0 \quad (3)$$

$$\mu \geq 0, \quad 4x + y \leq 20, \quad \mu(20 - 4x - y) = 0 \quad (4)$$

A multiplier is a measure of the change in the objective function when the constraint is relaxed.

So λ is the shadow price of capital, and μ is the shadow price of labour.

- (iii) • Neither constraint binds

$\lambda = \mu = 0$, which is inconsistent with (1) and (2).

- Only capital constraint binds

$\mu = 0$, so FOCs become $x^{-1} = \lambda$, $3y^{-1} = 4\lambda$, and so $x = 4y/3$.

Subst. into binding constraint leads to $x = 5$, $y = 15/4$.

Check other constraint: $4x + y = 20 + 15/4 > 20$, so not a solution.

- Only labour constraint binds

$\lambda = 0$, so FOCs become $x^{-1} = 4\mu$, $3y^{-1} = \mu$, and so $x = y/12$.

Subst. into binding constraint leads to $x = 5/4$, $y = 15$.

Check other constraint: $x + 4y = 5/4 + 60 > 20$, so not a solution.

- Both constraint binds

$x = y = 4$, so FOCs become $x^{-1} = \lambda + 4\mu$, $3y^{-1} = 4\lambda + \mu$,

leading to $\lambda = 11/60 \geq 0$, $\mu = 1/60 \geq 0$.

So solution is $x = y = 4$, along with these values of the multipliers.

- (iv) We are maximizing a concave function subject to linear constraints, so the Kuhn-Tucker conditions are sufficient, as well as necessary.

Hence the solution is optimal.

- (v) Objective function is no longer concave (see the function g from earlier part), but it is equivalent to one, so K-T FOCs are still sufficient.

Same optimal x and y but different λ and μ (shadow prices).

[65%]

3. (a) Schematically, the Independence Axiom states that $L_1 \succsim L_2 \iff L'_1 \succsim L'_2$
(where L' denotes compounding with a fixed third lottery).

Let $L_1 \succ L_2$. Then $L_1 \succsim L_2$, so $L'_1 \succsim L'_2$.

If it were the case that $L'_1 \sim L'_2$, then $L'_2 \succsim L'_1$, and so $L_2 \succsim L_1$ – a contradiction.

So $L'_1 \succ L'_2$.

(Now rerun this argument with the roles of L and L' reversed.)

Now let $L_1 \sim L_2$. Then $L_1 \succsim L_2$ and $L_2 \succsim L_1$, so $L'_1 \succsim L'_2$ and $L'_2 \succsim L'_1$.

Together, these last two relations imply that $L'_1 \sim L'_2$.

(Again, rerun this argument with the roles of L and L' reversed.)

$$L_1 \succsim L_2 \implies p \circ L_1 + (1-p) \circ L_3 \succsim p \circ L_2 + (1-p) \circ L_3$$

$$L_3 \succsim L_4 \implies q \circ L_3 + (1-q) \circ L_2 \succsim q \circ L_4 + (1-q) \circ L_2$$

$$\text{Take } q = 1-p \text{ and combine to give } p \circ L_1 + (1-p) \circ L_3 \succsim p \circ L_2 + (1-p) \circ L_4.$$

[40%]

- (b) When the possible outcomes are $X = \{-2, -1, 0, +1, +2\}$, and the lotteries are
 $L_1 = (1/5, 1/5, 1/5, 1/5, 1/5)$ and $L_2 = (1/15, 7/15, 0, 5/15, 2/15)$,
the common sufficient condition on the CDFs crossing once does not hold.

Start with L_2 ; compound the node that results in the outcome -1 with a lottery
 $(\frac{2}{7}, -1; \frac{3}{7}, 0; \frac{2}{7}, +1)$, and also compound the node that results in the outcome $+1$ with
a lottery $(\frac{1}{5}, -1; \frac{3}{5}, 0; \frac{1}{5}, +1)$;

this results in a lottery whose reduced version is L_1 .

So, starting with L_2 , to each of two possible outcomes x we have added a zero-mean
noise term $\tilde{z}$ distributed according to H_x ,

and L_1 is the reduced lottery of this compound lottery.

So L_1 is a mean-preserving spread of L_2 ,

and a risk-averse expected utility maximizer prefers L_2 to L_1 .

[20%]

- (c) (i) For 1SD, need CDF always below (necessary & sufficient condition),
so $p_1 \leq 1/3$, $p_3 \geq 1/3$, at least one strict (else same distribution), and so $p_1 < p_3$.
For LR non-monotonic, need either $p_1 > p_2$ or $p_2 > p_3$, such as
- $p_1 = 3/12, p_2 = 2/12, p_3 = 7/12$
 - $p_1 = 1/12, p_2 = 6/12, p_3 = 5/12$
- (ii) If LR is monotonically increasing then $p_1 \leq p_2 \leq p_3$,
and so $p_1 > 1/3$ is not possible (else $\sum p_k > 1$), nor is $p_3 < 1/3$ (else $\sum p_k < 1$).
So $p_1 \leq 1/3$, $p_3 \geq 1/3$, at least one strict (else same distribution), therefore
first-order stochastic dominance.

[40%]

4. Write v_k for $u(w_k)$.

(a) When both effort and the amount of output are observable ...

Participation constraint will bind:

$$e_1 : v = 2, w = 4, E[\pi | e_1] = \frac{3}{4} \cdot 40 + \frac{1}{4} \cdot 4 = 31, \text{ net } \pi = 27$$

$$e_0 : v = 0, w = 0, E[\pi | e_0] = \frac{1}{4} \cdot 40 + \frac{3}{4} \cdot 4 = 13, \text{ net } \pi = 13$$

$27 > 13$, so optimal contract is $(e = e_1, w = 4)$.

[15%]

Now effort is unobservable ...

(b) To induce e_0 , contract is same as full-info.: $(w_L = 0, w_H = 0)$.

To induce e_1 :

$$\text{Participation: } \frac{3}{4}v_H + \frac{1}{4}v_L - 2 \geq 0$$

$$\text{Incentive: } \frac{3}{4}v_H + \frac{1}{4}v_L - 2 \geq \frac{1}{4}v_H + \frac{3}{4}v_L - 0$$

If PC binds, then IC $\implies 0 \geq v_H + 3v_L$ — not poss.

Therefore only IC binds: $\frac{1}{2}v_H - \frac{1}{2}v_L = 2, v_H - v_L = 4$.

So set $v_L = 0$, & then $v_H = 4$, giving $w_L = 0, w_H = 16$.

$$E[w | e_1] = \frac{3}{4} \cdot 16 + \frac{1}{4} \cdot 0 = 12.$$

$$E[\pi - w | e_1] = 31 - 12 = 19.$$

$19 > 13$ (i.e. $E[\pi - w | e_0]$ from part (a)), so optimal contract is $(w_L = 0, w_H = 16)$.

[35%]

(c) • M/c Bad: output always 4, so $e = e_0, w = 0$, net profit is 4.

• M/c not Bad: $\Pr(M) = 2/3, \Pr(G) = 1/3$.

To induce e_0 : $(w_L = 0, w_H = 0), E[\pi | e_0] = \frac{1}{3} \cdot 40 + \frac{2}{3} \cdot 4 = 16$.

To induce e_1 : — output always 40

$$\text{Participation: } v_H - 2 \geq 0$$

$$\text{Incentive: } v_H - 2 \geq \frac{1}{3}v_H + \frac{2}{3}v_L - 0$$

Again, only IC binds: $\frac{2}{3}v_H - \frac{2}{3}v_L = 2, v_H - v_L = 3$.

Set $v_L = 0$, & then $v_H = 3$, giving $w_L = 0, w_H = 9$.

$$E[w | e_1] = 9 \text{ — } w \text{ always } w_H.$$

$$E[\pi - w | e_1] = 40 - 9 = 31 \text{ — output always high.}$$

$31 > 16$, so optimal contract is $(w_L = 0, w_H = 9)$.

$$\therefore E[\text{net } \pi | \text{inspection}] = \frac{1}{4} \cdot 4 + \frac{3}{4} \cdot 31 = 24\frac{1}{4}.$$

Now $E[\text{net } \pi | \text{no inspection}] = 19$, from part (b),

so prepared to pay up to $24\frac{1}{4} - 19 = 5\frac{1}{4}$ for the inspection.

[50%]

5. (a) The Walrasian demands maximize $\mathcal{L}(x_1, x_2, m) = \frac{x_1 x_2}{x_1 + x_2} - \lambda(p_1 x_1 + p_2 x_2 - m)$. The first order conditions are given by:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \frac{x_2}{x_1 + x_2} - \frac{x_1 x_2}{(x_1 + x_2)^2} - \lambda p_1 = 0$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = \frac{x_1}{x_1 + x_2} - \frac{x_1 x_2}{(x_1 + x_2)^2} - \lambda p_2 = 0$$

$p_1 x_1 + p_2 x_2 = m$ (budget constraint binds as utility is increasing),

which imply: $p_1 x_1^2 = p_2 x_2^2$. Together with $p_1 x_1 + p_2 x_2 = m$ we find:

$$x_1(p_1, p_2, m) = \frac{m}{\sqrt{p_1}(\sqrt{p_1} + \sqrt{p_2})} \quad x_2(p_1, p_2, m) = \frac{m}{\sqrt{p_2}(\sqrt{p_1} + \sqrt{p_2})}$$

There should be some mention to second order conditions, quasi-concavity etc. (Don't take points out if they don't check the Hessian, but they should acknowledge it)

Clearly they are linear in income. ($x_i(p_1, p_2, m) = m x_i(p_1, p_2, 1)$).

It is homogeneous of degree 0: $x_i(tp, tm) = \frac{tm}{\sqrt{tp_1}(\sqrt{tp_1} + \sqrt{tp_2})} = \frac{m}{\sqrt{p_1}(\sqrt{p_1} + \sqrt{p_2})} = x_i(p, m)$.

(b)

$$\begin{aligned} (1, p) \cdot Z(1, p') &= \left(\sum_a \frac{\omega_1^a + p' \omega_2^a}{1 + \sqrt{p'}} - \sum_a \omega_1^a \right) + p \left(\sum_a \frac{\omega_1^a + p' \omega_2^a}{\sqrt{p'}(1 + \sqrt{p'})} - \sum_a \omega_2^a \right) \\ &= \left(\frac{\omega_1 + p' \omega_2}{1 + \sqrt{p'}} - \omega_1 \right) + p \left(\frac{\omega_1 + p' \omega_2}{\sqrt{p'}(1 + \sqrt{p'})} - \omega_2 \right) \\ &= (\omega_1 + p' \omega_2) \left(\frac{\sqrt{p'} + p}{\sqrt{p'}(1 + \sqrt{p'})} \right) - (\omega_1 + p \omega_2) \end{aligned}$$

and the equivalence follows directly.

- (c) The WARP is satisfied if whenever $(1, p) \cdot Z(1, p') \leq 0$, we have that $(1, p') \cdot Z(1, p) > 0$ (or $\omega_1 + p \omega_2 > (\omega_1 + p' \omega_2) \frac{\sqrt{p} + p}{\sqrt{p} + p'}$). Suppose that $(1, p) \cdot Z(1, p') \leq 0$, then by (1) we have that $\omega_1 + p \omega_2 \geq (\omega_1 + p' \omega_2) \frac{\sqrt{p'} + p}{\sqrt{p'} + p'} > (\omega_1 + p' \omega_2) \frac{\sqrt{p} + p}{\sqrt{p} + p'}$, where the strict inequality follows because

$$\begin{aligned} \frac{\sqrt{p'} + p}{\sqrt{p'} + p'} &> \frac{\sqrt{p} + p}{\sqrt{p} + p'} \\ \Leftrightarrow (\sqrt{p'} + p)(\sqrt{p} + p') &> (\sqrt{p} + p)(\sqrt{p'} + p') \\ \Leftrightarrow p'(\sqrt{p'} - \sqrt{p}) &> p(\sqrt{p'} - \sqrt{p}) \end{aligned}$$

which is always satisfied. (If $p' > p$ the common factor does not alter the sign, and if $p' < p$ we can cancel the common factor by changing the sign.) In particular this implies that the Walrasian equilibrium will be unique and stable.

Some students might argue that the Walrasian demands are linear in income, and hence the aggregate demand can be interpreted as the demand of an individual that has the aggregate income in the economy. Since individual demands satisfy the WARP so do the aggregate ones.

- (d) To find the Walrasian equilibrium we equate the aggregate demand to 0 in one of the markets (given Walras' law, the other market will also clear).

$\frac{\omega_1 + p^* \omega_2}{1 + \sqrt{p^*}} = \omega_1$ implies $\sqrt{p^*} = \frac{\omega_1}{\omega_2}$ so $p^* = \left(\frac{\omega_1}{\omega_2} \right)^2$. As the relative aggregate endowment of good 1 increases its relative price decrease (or the relative price of good 2 increases).

6. (a) Denoting by q the price of α and z^A the portfolio demand for agent A , the budget constraint for the agent A is $x_0 = 1 - qz^A$, $x_1 = 1$, $x_2 = z^A$. Substituting in the utility function, the agent chooses z^A to maximize $2 \ln(1 - qz^A) + \ln(1) + \ln(z^A)$. The FOC is $\frac{-2q}{1 - qz^A} + \frac{1}{z^A} = 0$, which implies $z^A = \frac{1}{3q}$.

For agent B , z^B maximize $2 \ln(-qz^B) + \ln(1) + \ln(1 + z^B)$. the FOC is $\frac{2}{z^B} + \frac{1}{1 + z^B} = 0$, which implies $z^B = -\frac{2}{3}$.

In equilibrium $z^A = -z^B$, hence $q^* = \frac{1}{2}$. The equilibrium allocations are: $x^A = (\frac{2}{3}, 1, \frac{2}{3})$, $x^B = (\frac{1}{3}, 1, \frac{1}{3})$.

The vector of marginal utilities for the agents are: $MU^A = (3, 1, \frac{3}{2})$, $MU^B = (6, 1, 3)$. These two vectors are not co-linear, hence the allocation is not Pareto efficient.

- (b) At the equilibrium, $u^B = 2 \ln \frac{1}{3} + \ln 1 + \ln \frac{1}{3} = 3 \ln \frac{1}{3}$. The constrained optimisation problem is maximizing the utility of A subject to agent B having utility $3 \ln \frac{1}{3}$ and given that with the financial assets available agents cannot exchange any consumption in state 1.

The constraint can be rewritten as $x^2(1 - y) = \frac{1}{27}$. The Lagrangian is given by $\mathcal{L} = 2 \ln(1 - x) + \ln(y) - \lambda(x^2(1 - y) - \frac{1}{27})$ and the first order conditions are:

$$-\frac{2}{1 - x} - 2\lambda x(1 - y) = 0, \quad \frac{1}{y} + \lambda x^2 = 0, \quad x^2(1 - y) - \frac{1}{27} = 0$$

Solving we the two first conditions we get $-2xy + 2(1 - y)(1 - x) = 0$ or $x = 1 - y$, which with the constraint gives $x = \frac{1}{3}$, $y = \frac{2}{3}$.

This implies that the utility of A is maximized when her consumption is $x_0 = 1 - x = \frac{2}{3}$, $x_1 = 1$, $x_2 = y = \frac{2}{3}$, which corresponds to the equilibrium allocation.

Hence the equilibrium allocation is constrained Pareto efficient.

- (c) We can solve for the state prices (or canonical asset prices) and then use the invariance theorem (that is quicker than solving with the given assets, but approaches are correct and should reach the same conclusion).

For state prices we solve the endowment economy. (Candidates can use the formula for Cobb-Douglas demands directly.) The demands of the two agents are:

$$x^A = (\frac{1}{2}(1 + p_1), \frac{1}{4p_1}(1 + p_1), \frac{1}{4p_2}(1 + p_1)), \quad x^B = (\frac{1}{2}(p_1 + p_2), \frac{1}{4p_1}(p_1 + p_2), \frac{1}{4p_2}(p_1 + p_2))$$

The market clearing conditions ($x_0^A + x_0^B = 1$, $x_1^A + x_1^B = 2$, $x_2^A + x_2^B = 1$) lead to $p_1 = \frac{1}{4}$ and $p_2 = \frac{1}{2}$.

The equilibrium allocations are $x^A = (\frac{5}{8}, \frac{5}{4}, \frac{5}{8})$ and $x^B = (\frac{3}{8}, \frac{3}{4}, \frac{3}{8})$.

The asset prices are $q_\beta = p_1 + p_2 = \frac{3}{4}$, and $q_\alpha = p_2 = \frac{1}{2}$.

The portfolios demands are $z_\alpha^A = \frac{3}{8}$, $z_\beta^A = \frac{1}{4}$, and $z^B = -z^A$.

The vectors of marginal utilities are $MU^A = (\frac{16}{5}, \frac{4}{5}, \frac{8}{5})$ and $MU^B = (\frac{16}{3}, \frac{4}{3}, \frac{8}{3})$ which are clearly collinear.

Hence the equilibrium allocations are Pareto efficient.

This is not surprising as markets are complete.